1. Posterior sampling: why we did it

The purpose of this short presentation is to explain our posterior sampling study. Our question is: how sure are we about the model behind that forecast? A Bayesian model does not give us only one set of predictor weights. It gives us a distribution of plausible weights after seeing the data. That is called the posterior. A weight describes how a predictor is associated with the final grade while the other predictors are held fixed.

Think of sampling as taking many plausible versions of the fitted model, each consistent with its assumptions and the observed data. We can then ask whether an association is usually positive, whether it is large enough to matter, and how the uncertainty affects a forecast. We are drawing possible parameter values, not inventing additional students.

This short section explains the simulation checks in our earlier consolidated study. That fit used 148 earlier students. The weekly model uses 153 training students and different predictors, so the numerical checks I show are not new validation results for the weekly model.

2. Two ways to sample the same posterior

There are two ways we sampled the same posterior in this study. First, this model has an exact mathematical solution. We can draw directly from it. We took 20,000 independent draws, meaning that one draw does not depend on the previous draw.

Second, we ran MCMC. The letters mean Markov chain Monte Carlo. In plain language, it is a sequence of random updates designed to explore the posterior. A chain is one run of those updates. Successive draws can be related, unlike the independent draws on the left.

We used a method called Gibbs sampling: update the predictor weights given the current noise level, then update the noise level given those weights, and repeat. Noise means variation in final grades that the predictors do not explain.

Why do both? The exact solution is a reference answer for this model. Comparing Gibbs sampling with it gives us a strong numerical check and helps us understand sampling before considering models without exact solutions. The weekly forecast still uses the exact formula. MCMC did not replace it or make its predictions automatically better.

3. How Gibbs MCMC works

Here is what the Gibbs sampler actually did. We began four separate runs at different noise levels, rather than relying on a single starting point. In each run we repeatedly sampled predictor weights and then the unexplained variance. If we run long enough and the procedure is working, the retained draws should explore the same posterior regardless of where we started.

We discarded the first 1,000 updates from each chain as warm-up, so the initial settings would have less influence on our summary. We then kept 4,000 draws from each of the four chains: 16,000 retained MCMC draws in total. These are separate from the 20,000 independent direct draws on the previous slide.

The recorded experiment sampled the final-grade regression conditional on a fixed set of predictors. Missing predictor values had been filled using training-data medians, and that filled-in design stayed fixed during sampling. Therefore this experiment does not account for uncertainty in those missing values, and it is not joint sampling of every unknown in the full DAG. That matters when interpreting how complete the reported uncertainty is.

4. The sampling agreed with the exact solution

We did not just run the sampler and assume it worked. The chart shows part of the retained chains for one predictor weight and the unexplained variance. The colored lines overlap and move around a common region. That is a useful visual check, but it is not enough on its own.

R-hat compares variation within and between chains. Values near one indicate that the runs are behaving similarly; the largest value here was 1.0007. Effective sample size accounts for the fact that linked MCMC draws repeat information. The smallest bulk effective sample size was 12,194. This is an effective number of parameter draws, not a number of students.

We also compared the sampled parameter means with the exact mathematical means. The largest difference was about 2.01 Monte Carlo standard errors: the estimated numerical uncertainty caused by taking a finite number of draws. Together, these checks support agreement with the exact posterior for this fitted model. They do not prove that the model is correct, that a predictor is causal, or that future grade forecasts will be accurate.

Source: consolidated presentation slide 5; posterior_study/analysis.py and posterior_results/provenance.json. The trace displays the first 800 retained updates per chain.

5. What the posterior tells us about the inputs

Now we can use the posterior to describe uncertainty about individual predictor weights. These are results from the earlier consolidated study: the final regression fit used 148 earlier students and the larger grade bundle. They are separate from the weekly model and its 153-student training fit.

Each estimate is in final-grade percentage points for a one-training-standard-deviation increase in that predictor, holding the other predictors fixed. For example, LMS page views have a posterior mean of plus 0.34 points, with a 95% credible interval from minus 0.21 to plus 0.88. Under this model, 95% of the posterior probability for that coefficient lies in that interval. Because zero lies inside it, the direction is uncertain at this interval level.

All six non-grade predictor intervals include zero. In this fit, the posterior probability that any one of these six weights has an absolute magnitude above one final-grade point is below 1% for each weight. This supports describing the conditional associations as small and uncertain under the stated model. It does not prove that wellness or engagement never matter, and it does not measure what would happen if we intervened to change them.

The coefficient summaries use the analytic posterior; simulation agreement validates the numerical sampling procedure. Source: consolidated presentation slide 9 and posterior_results/coefficient_posteriors.csv.

6. What this establishes—and what it leaves open

The practical result is confidence in the posterior computation, within the assumptions of that earlier model. It also gives us a clearer language for uncertainty. A credible interval for a predictor weight describes uncertainty about an association. A prediction interval describes uncertainty about one student's eventual outcome, including unexplained student-to-student variation. Those intervals answer different questions, and we should not read them interchangeably.

The weekly model can calculate both kinds of quantities using its exact formulas, so it does not need MCMC to make forecasts. Sampling is useful here as a computational check and a way to understand the distribution, rather than a new source of predictive signal.

To show that a source of data improves forecasts, we still need comparisons on later students. To show that a coaching action changes grades, we need a causal study. Neither follows from well-behaved sampling.

To close: we checked that simulation reproduces the exact posterior and used that posterior to describe uncertain conditional associations. Our next evidence for practical forecasting comes from later-student prediction tests, which are a separate analysis. The earlier study conditions on training-median-filled inputs, so its intervals omit uncertainty in those missing values. Good chain diagnostics cannot repair missing-data assumptions, establish grade-release timing, prove a coaching effect, or validate a different fitted model.

References: consolidated presentation slides 4, 5, 9 and 17; posterior_study/analysis.py; posterior_results/provenance.json; Vehtari et al. (2021), improved R-hat and effective sample size diagnostics, DOI 10.1214/20-BA1221.